



**KCG**  
**COLLEGE OF TECHNOLOGY**  
AFFILIATED TO ANNA UNIVERSITY | AUTONOMOUS

# **COURSE MATERIAL**

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**FOR**

**GE3752 TOTAL QUALITY MANAGEMENT COMMON TO  
CIVIL/AUTOMOBILE/ FASHION TECHNOLOGY/IT /CSE**

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by,

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## SYLLABUS

### Unit I Introduction to TQM

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Introduction - Need for quality - Evolution of quality - Definitions of quality - Dimensions of product and service quality - Basic concepts of TQM - TQM Framework - Contributions of Deming, Juran and Crosby - Barriers to TQM - Quality statements - Customer focus - Customer orientation, Customer satisfaction, Customer complaints, Customer retention - Costs of quality.

### Unit II TQM Principles

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Leadership - Strategic quality planning, Quality Councils - Employee involvement - Motivation, Empowerment, Team and Teamwork, Quality circles Recognition and Reward, Performance appraisal - Continuous process improvement - PDCA cycle, 5S, Kaizen - Supplier partnership - Partnering, Supplier selection, Supplier Rating

### Unit III TQM TOOLS AND TECHNIQUES I

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The seven traditional tools of quality - New management tools - Six sigma: Concepts, Methodology, applications to manufacturing, service sector including IT - Bench marking - Reason to bench mark, Bench marking process - FMEA - Stages, Types

### Unit IV TQM TOOLS AND TECHNIQUES II

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Control Charts - Process Capability - Concepts of Six Sigma - Quality Function Development (QFD) - Taguchi quality loss function - TPM - Concepts, improvement needs  
- Performance measures

## Unit V QUALITY SYSTEMS

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Need for ISO 9000 - ISO 9001-2008 Quality System - Elements, Documentation, Quality Auditing - QS 9000 - ISO 14000 - Concepts, Requirements and Benefits - TQM Implementation in manufacturing and service sectors

### **E. Content Beyond Syllabus**

Six sigma of various industries.

### **TEXTBOOKS:**

1. Dale H. Besterfield, et al., "Total quality Management", Pearson Education Asia, Third Edition, Indian Reprint (2006)... (UNIT I, II, III, IV, V)

### **REFERENCES:**

1. James R. Evans and William M. Lindsay, "The Management and Control of Quality", 8th Edition, First Indian Edition, Cengage Learning, 2012.
2. Suganthi.L and Anand Samuel, "Total Quality Management", Prentice Hall (India) Pvt. Ltd., 2006.
3. Janakiraman. B and Gopal .R.K., "Total Quality Management - Text and Cases", Prentice Hall (India) Pvt. Ltd., 2006.

## UNIT – I

### TOTAL QUALITY MANAGEMENT – INTRODUCTION

#### 1.1 Introduction to Total Quality Management

Total Quality Management (TQM) is a comprehensive management philosophy aimed at continuously improving the quality of products, services, and processes in an organization. It represents an enhancement of the traditional approach to management by emphasizing customer satisfaction, employee involvement, and continuous improvement.

The term *Total* signifies that quality involves every individual and every activity within the organization. **Quality** refers to the degree of excellence achieved by a product or service in meeting customer requirements. **Management** involves planning, organizing, directing, and controlling resources effectively. Thus, TQM integrates people, processes, and systems to achieve organizational excellence.

TQM can be defined as the application of quantitative methods and effective utilization of human resources to improve all organizational processes and exceed customer expectations both now and in the future.

#### 1.2 Concept and Definition of Quality

Quality is a relative and customer-oriented concept. It can be expressed quantitatively using the relationship:

$$Q = \frac{P}{E} \quad \text{or} \quad Q = PE$$

where **Q** denotes Quality, **P** represents Performance, and **E** stands for Expectation. According to this relationship, quality improves when performance increases or when customer expectations are properly understood and satisfied. Hence, quality is largely determined by customer perception.

#### 1.3 Dimensions of Quality

Quality cannot be described by a single attribute and is instead characterized by multiple dimensions. Performance refers to the primary operating characteristics of a product. Features are the secondary characteristics that enhance product appeal. Conformance indicates adherence to specifications and standards. Reliability refers to consistency of performance over time, while durability represents the useful life of the product including repair. Service involves the ease of handling complaints and

repairs, and response refers to the quality of human interaction such as courtesy and promptness. Aesthetics relate to sensory characteristics like appearance and finish, whereas reputation reflects past performance and brand image.

## 1.4 Quality Planning

Quality planning is a systematic process that ensures customer requirements are translated into product and process specifications.

### 1.4.1 Steps in Quality Planning

Quality planning begins with the establishment of quality goals aligned with organizational objectives. This is followed by identifying customers and understanding their needs. Based on customer requirements, product features are developed, and suitable process features are designed. Finally, process controls are established and transferred to operational levels to ensure consistent quality performance.

### 1.4.2 Important Considerations in Quality Planning

Organizations with superior quality and larger market share generally achieve higher profitability. Quality and market share independently influence profitability. Therefore, quality planning should be based on customer needs rather than merely meeting specifications. For standardized products, planning should aim for perfection, whereas for value-oriented products, quality planning should balance performance and cost.

## 1.5 Quality Costs

Quality costs represent the costs incurred due to the failure to achieve required quality standards. In simple terms, quality cost is the cost of poor quality.

### 1.5.1 Elements of Quality Cost

Quality costs are classified into prevention costs, appraisal costs, internal failure costs, and external failure costs.

### 1.6 Prevention Costs

Prevention costs are incurred to avoid defects before they occur. These include costs related to market research, customer analysis, product and service design, purchasing control, process planning, employee training, and administrative quality planning activities.

## 1.7 Appraisal Costs

Appraisal costs are associated with evaluating products and services to ensure they meet quality standards. These include inspection and testing of incoming materials, in-process inspection, final product inspection, external audits, and review of inspection and test data.

## 1.8 Internal Failure Costs

Internal failure costs arise when defects are detected before the product reaches the customer. These include costs associated with design failures, purchasing errors, rework, scrap, re-inspection, and process inefficiencies.

## 1.9 External Failure Costs

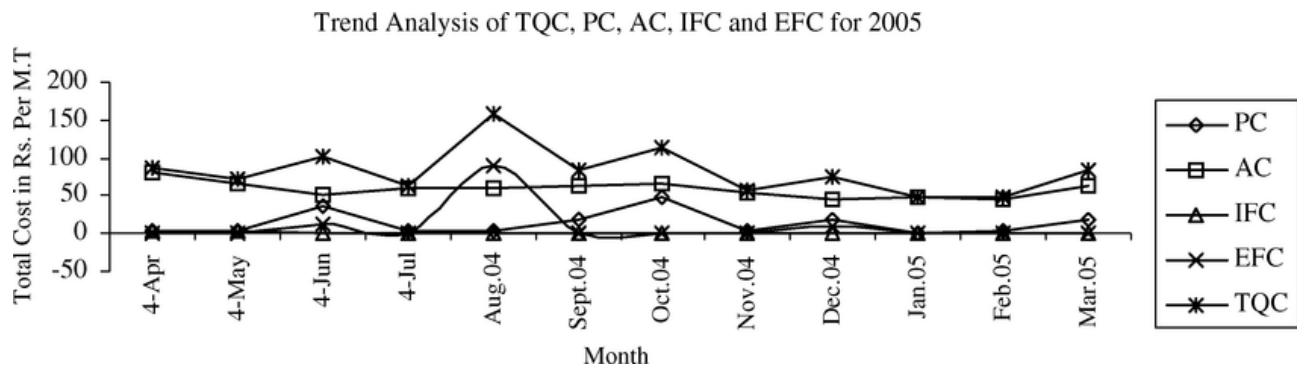
External failure costs occur when defects are identified after the product has been delivered to customers. These include costs related to complaint handling, returned goods, recall and retrofit activities, warranty claims, legal liabilities, penalties, loss of goodwill, and loss of future sales.

## 1.10 Analysis of Quality Costs

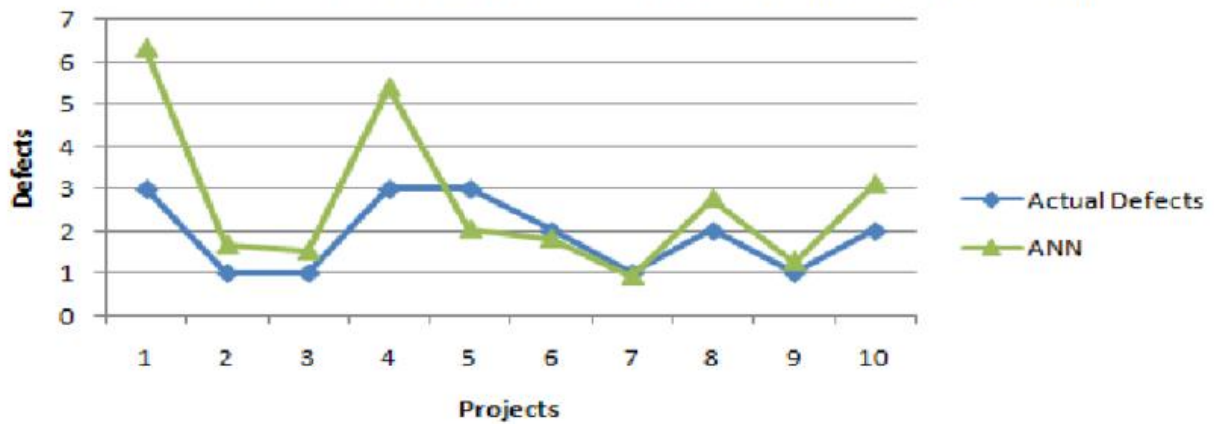
The purpose of quality cost analysis is to determine the cost of maintaining a specific level of quality. It provides feedback to management on the effectiveness of quality assurance systems and assists in identifying opportunities for improvement.

### 1.10.1 Trend Analysis

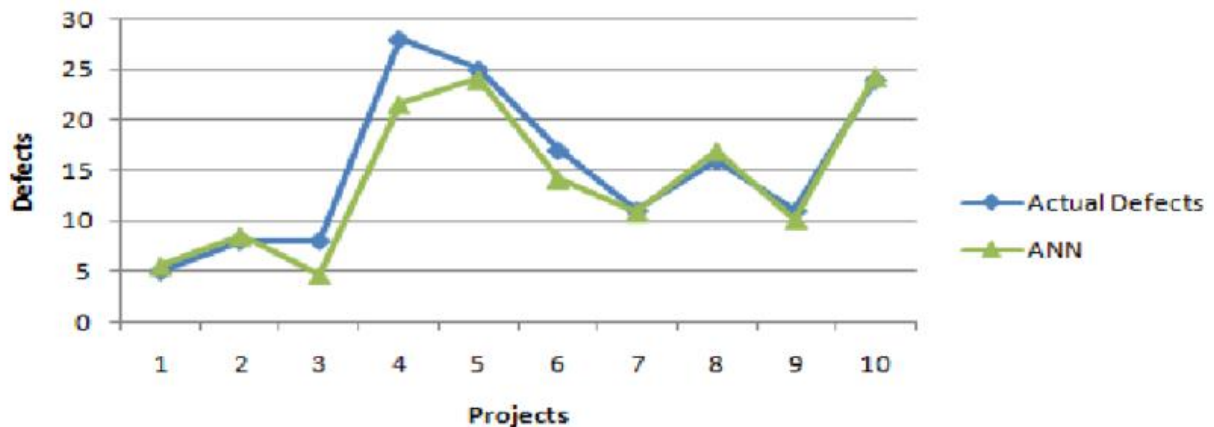
Trend analysis involves comparing quality-related data over different time periods to identify patterns and variations. Graphical representations such as line charts are effective tools for presenting trends in quality costs, defect rates, or performance indicators.



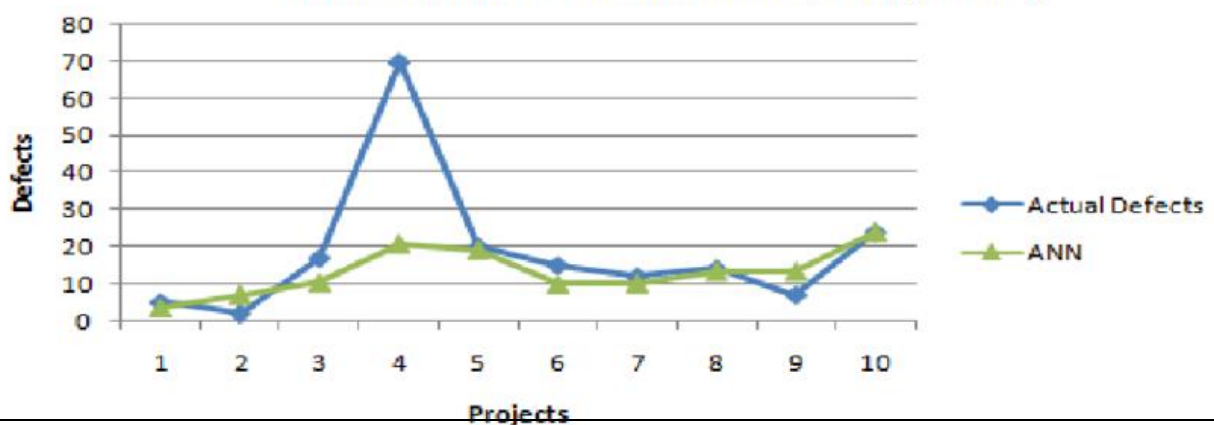
**Comparison of defects data for 10 Projects during Validation :  
ANN Prediction VS Actual Defects ( Requirement Phase )**

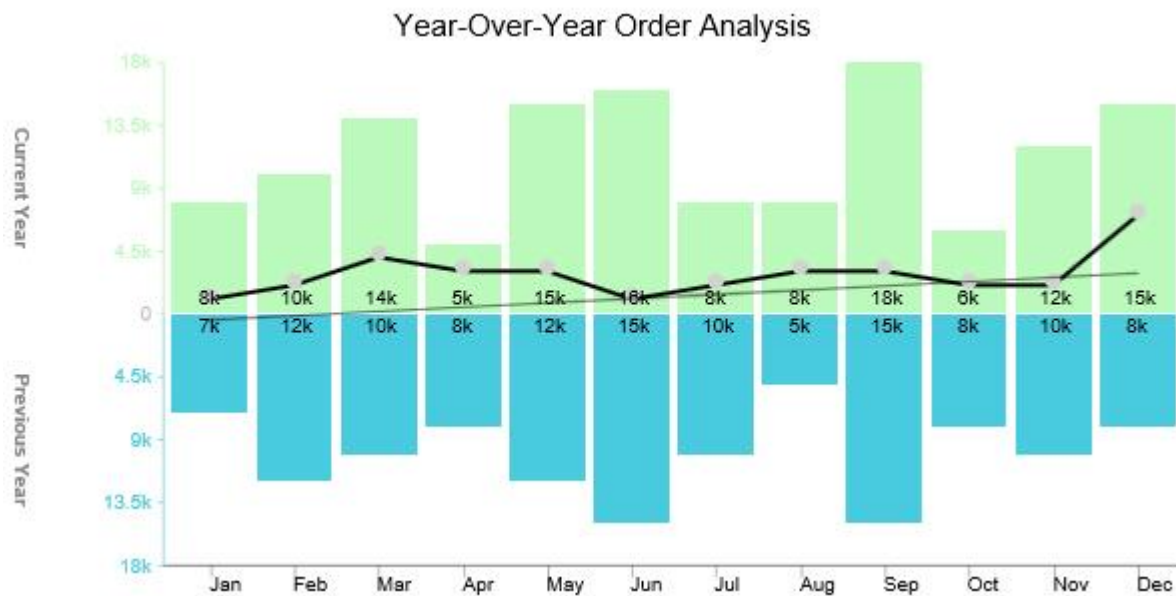


**Comparison of defects data for 10 Projects during Validation : ANN  
Prediction VS Actual Defects ( Construction Phase )**



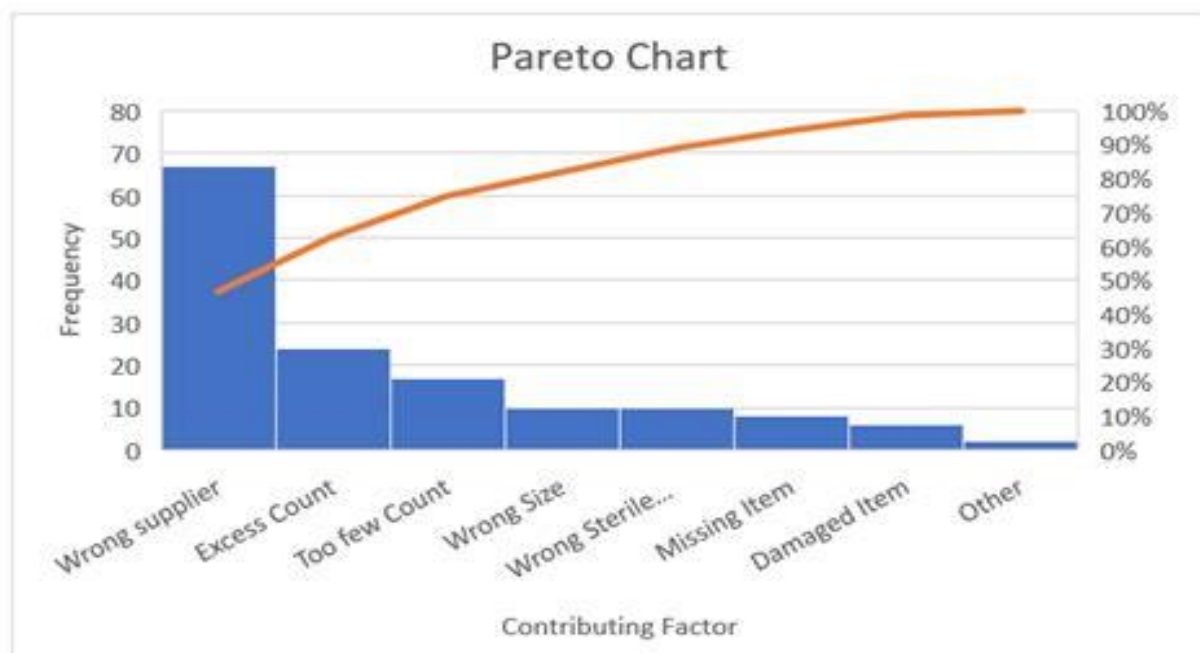
**Comparison of defects data for 10 Projects during Validation :  
ANN Prediction VS Actual Defects ( Testing Phase )**





### 1.10.2 Pareto Analysis

Pareto analysis is based on the principle proposed by Joseph Juran, which states that a small number of causes are responsible for the majority of quality problems. By identifying and addressing these vital few causes, organizations can achieve significant improvement with minimal effort.





### 1.11 Basic Concepts of Total Quality Management

The basic concepts of TQM include strong top management commitment, customer focus, process orientation, continuous performance measurement, employee involvement and empowerment, continuous improvement, benchmarking, teamwork, supplier collaboration, training, effective inventory management, open communication, and monitoring of quality costs.

### 1.12 Pillars of Total Quality Management

TQM is supported by four major pillars: disciplined problem solving, effective interpersonal skills, teamwork, and a structured quality improvement process.

### 1.13 Principles of Total Quality Management

The principles of TQM emphasize meeting customer requirements correctly the first time and every time. There must be clearly agreed requirements for both internal and external customers. Employees at all levels should be involved, two-way communication must be encouraged, continuous training should be provided, and top management must actively participate. Waste elimination, prevention of defects, creativity promotion, performance measurement, and teamwork are essential for successful TQM implementation.

### 1.14 Six Basic Concepts of TQM

The six fundamental concepts of TQM are management commitment, customer focus, total workforce involvement, continuous improvement, treating suppliers as partners, and establishing performance measures for processes.

### 1.15 Obstacles in Implementing TQM

Organizations may face several obstacles while implementing TQM, such as lack of management commitment, resistance to cultural change, improper planning, inadequate training, incompatible organizational structures, ineffective measurement systems, insufficient focus on customers, poor empowerment practices, and failure to pursue continuous improvement.

### 1.16 Benefits of Total Quality Management

Effective implementation of TQM results in improved quality, increased employee participation, stronger teamwork, better working relationships, enhanced customer satisfaction, improved employee morale, higher productivity, effective communication, increased profitability, and improved market share.

### 1.17 Leadership in Total Quality Management

Leadership plays a vital role in TQM. Leaders shape, promote, protect, and exemplify organizational values.

### 1.17.1 Roles of Leaders

Leaders perform various roles such as producer, director, coordinator, checker, stimulator, mentor, innovator, and negotiator.

### 1.17.2 Characteristics of Quality Leaders

Quality leaders focus on customer needs, empower employees, emphasize prevention and improvement, promote collaboration, train and coach employees, learn from problems, continuously improve communication, demonstrate commitment to quality, select suppliers based on quality, establish supportive systems, and recognize team effort.

### 1.18 The Seven Habits of Highly Effective People

Highly effective individuals are proactive, begin with the end in mind, prioritize important tasks, think in terms of win-win outcomes, seek to understand before being understood, work synergistically, and continuously renew their abilities.

### 1.19 Role of Senior Management in TQM

Senior management supports TQM through management by wandering around, strategic problem solving, maintaining strong information systems, implementing recognition and reward mechanisms, dedicating time to quality initiatives, ensuring effective communication, identifying talent, acting as role models, removing barriers, promoting training, establishing quality policies, and leading cultural transformation.

### 1.20 Quality Council

A quality council provides overall direction for TQM initiatives. It consists of the chief executive officer, senior managers, a coordinator or consultant, and a union representative.

#### 1.20.1 Functions of Quality Council

The quality council develops vision, mission, and quality policy statements, prepares strategic and annual quality plans, designs training programs, monitors the cost of poor quality, establishes performance measures, identifies improvement projects, forms cross-functional teams, and revises recognition and reward systems.

### 1.21 Quality Statements

A vision statement describes what the organization aspires to become in the future. A mission statement explains who the customers are, what the organization does, and how it operates. A quality policy statement guides employees in delivering products and services that meet or exceed customer expectations through continuous improvement.

## 1.22 Strategic Quality Planning

Strategic quality planning distinguishes long-term goals from short-term objectives. Goals should improve customer satisfaction, employee satisfaction, and process performance. They must be measurable, data-driven, time-bound, achievable, and challenging.

### 1.22.1 Steps in Strategic Quality Planning

Strategic quality planning involves identifying customer needs, positioning the organization, predicting future conditions, conducting gap analysis, closing identified gaps, aligning resources, and implementing improvement plans.

## 1.23 Implementation of Total Quality Management

TQM implementation begins with strong management commitment and leadership. Organizations must define their aims, adopt a new philosophy, understand the purpose of inspection, avoid price-based purchasing, continuously improve systems, provide training, establish leadership, eliminate fear, promote teamwork, remove barriers to workmanship pride, encourage education and self-improvement, and take action to achieve organizational transformation.

